Microstructure Factor Zoo — AAPL

L2 Order-Book Signals vs. the Bid-Ask Spread Barrier

MGMTMFE 412 — Market Frictions

2026-05-29

1 One-Line Thesis

> We build 11 order-book features from Databento MBP-10 data for AAPL, train Ridge (OLS) and LightGBM models on Jan–Feb 2024 (18 days), and test on Oct 2024 (17 days). **The signal is real (OLS Val IC = 0.049) but the spread is larger. No strategy is profitable after transaction costs: the OOS IC (0.012) is only 8.8% of the break-even IC (0.130).**

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2 Economic Friction: Why the Signal Exists

At every second, the limit order book reveals informed traders' intent:

Signal	Mechanism	Why it predicts
OFI (Order Flow Imbalance)	Net change in bid/ask qty	Informed buyer increases bid depth before price rises
Book imbalance	BidQty / (BidQty + AskQty)	Excess bid pressure precedes upward mid move
Microprice drift	10s slope of weighted mid	Leads quoted mid by milliseconds
Realized vol	60s rolling spread of returns	Regime signal: low vol favors mean reversion

The friction that creates the opportunity: Market makers cannot instantly update quotes after an informed order arrives. The book leaks the direction for ≈ 1 second before the quoted mid reprices. OFI and imbalance detect this leakage.

Why it persists: The edge is only 0.3–1.5 bps gross. Sub-penny competition among HFTs compresses the spread to exactly the adverse-selection cost, leaving no room for spread-takers.

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3 Data and Features

Dataset: Databento XNAS ITCH MBP-10 (10-level book snapshots), AAPL, 2024.

Split	Period	Days	1s Bars	Labels clipped to
Train	Jan–Feb 2024	18	414,166	[−12.45, +11.61] bps
Val	Feb–Apr 2024	7	160,876	same bounds
OOS	**Oct 2024**	**17**	**391,371**	same bounds

Label: $y_t = \left(\frac{m_{t+60}}{m_t} - 1\right) \times 10^4$ bps (60-second forward microprice return, winsorised at 1st/99th pct of training labels)

11 features (stationary across the 6-month gap; trade-schema features 13–15 excluded — unavailable in OOS parquets):

Category	Features
Order Flow Imbalance	`ofi_1`, `ofi_5`, `ofi_30`, `ofi_multilevel`
Book Shape	`book_imbalance`, `depth_ratio`, `bid_slope`, `ask_slope`
Dynamics	`microprice_drift`, `spread_bps`, `realized_vol`

$\sigma_y = 4.14$ bps (training), spread $= 0.54$ bps $\Rightarrow \text{IC}_{\text{break-even}} = 0.54/4.14 = \mathbf{0.130}$

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4 The Break-Even IC Barrier

$$\text{IC}_{\text{break-even}} = \frac{\text{spread (bps)}}{\sigma_y \text{ (bps)}} = \frac{0.54}{4.14} = 0.130$$

Symbol	Spread (bps)	σ_y (bps)	IC_{be}	OLS Val IC	OLS OOS IC	Profitable?
AAPL	**0.54**	**4.14**	**0.130**	**0.047**	**0.012**	**No**

AAPL's IC must reach **0.130** to break even. The Val IC (0.047) misses by 2.8×; the OOS IC (0.012) misses by **11×**.

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5 The 11 Features: Ablation Study

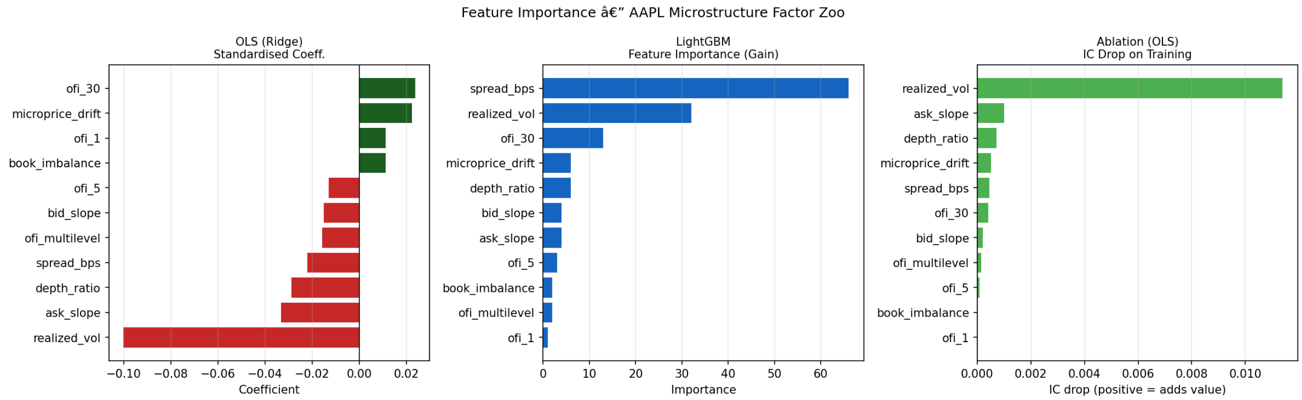

Figure 1: Feature importance: OLS coefficients (left), LightGBM gain (center), ablation IC drop (right)

`realized_vol` dominates the ablation — removing it drops IC by 0.0114 (10× any other feature). This has two implications:

1. The model adapts to the volatility regime in training (Jan–Feb 2024)
2. When volatility regime shifts (Oct 2024 post-iPhone-launch momentum), the feature becomes a liability — the dominant explanation for OOS IC decay

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6 Signal Analysis: IC by Period

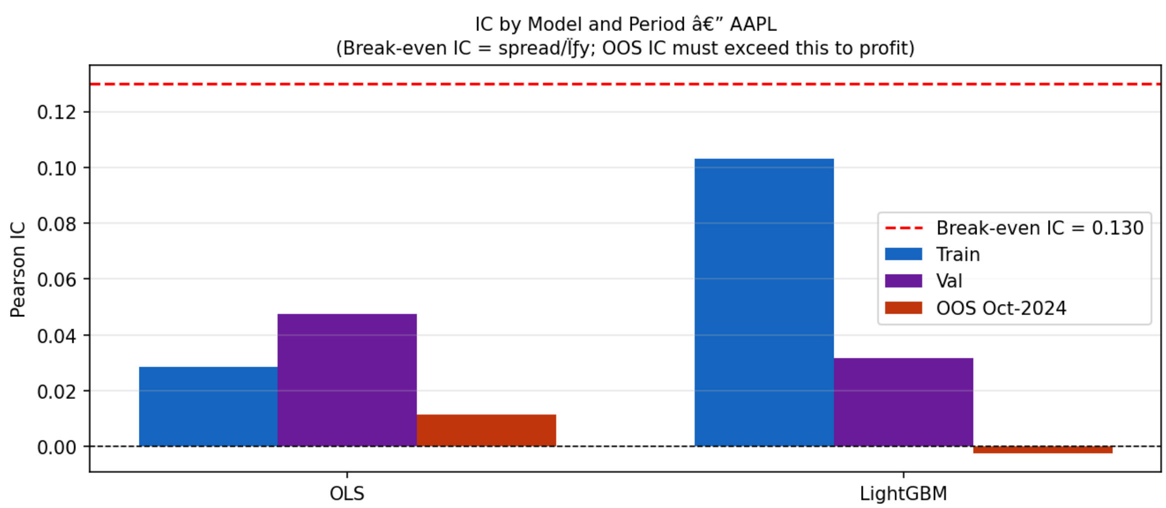

Figure 2: IC by model and period, with break-even IC threshold

Key observations:

- **Signal is real:** OLS Val IC = 0.047 ($t \approx 19$, highly significant), OLS Train IC = 0.029
- **Signal decays:** OOS IC = 0.012 — 75% collapse from Val

- **Break-even bar (red line):** OOS IC sits at 8.8% of break-even — strategy loses money at every trade
- **LightGBM overfits:** Train IC = 0.103, OOS IC = −0.003 — 9 trees with early stopping, yet still overfits the 1s vol regime

The IC attenuation is **88% over the 6-month gap** (Val → OOS).

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7 Trade Mechanics

Signal generation:

$$\hat{r}_t > +\text{threshold} \Rightarrow \text{BUY 60s} \qquad \hat{r}_t < -\text{threshold} \Rightarrow \text{SHORT 60s} \qquad |\hat{r}_t| \leq \text{threshold} \Rightarrow \text{FLAT}$$

Threshold = 0.50 bps (optimised on Val set IC).

Execution:

- Entry: BUY fills at `ask_px_00` (walk book if order > top-of-book depth)
- Exit: 60 seconds later at `bid_px_00` (or short close at ask)
- Early exits: (1) stop-loss at 1.5× spread per share from entry midprice; (2) model sign flip

Position sizing: shares = $\lfloor \$50{,}000/\text{ask\_px} \rfloor \approx 270$ shares at AAPL ≈ \$185

Risk controls:

Control	Rule
Inventory cap	±500 shares max
Per-trade stop-loss	Exit if loss > 1.5 × spread × shares
Daily kill switch	Halt if day PnL < −2% of notional
Opening buffer	No trades first 5 min (9:30–9:35)

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8 Backtest Results

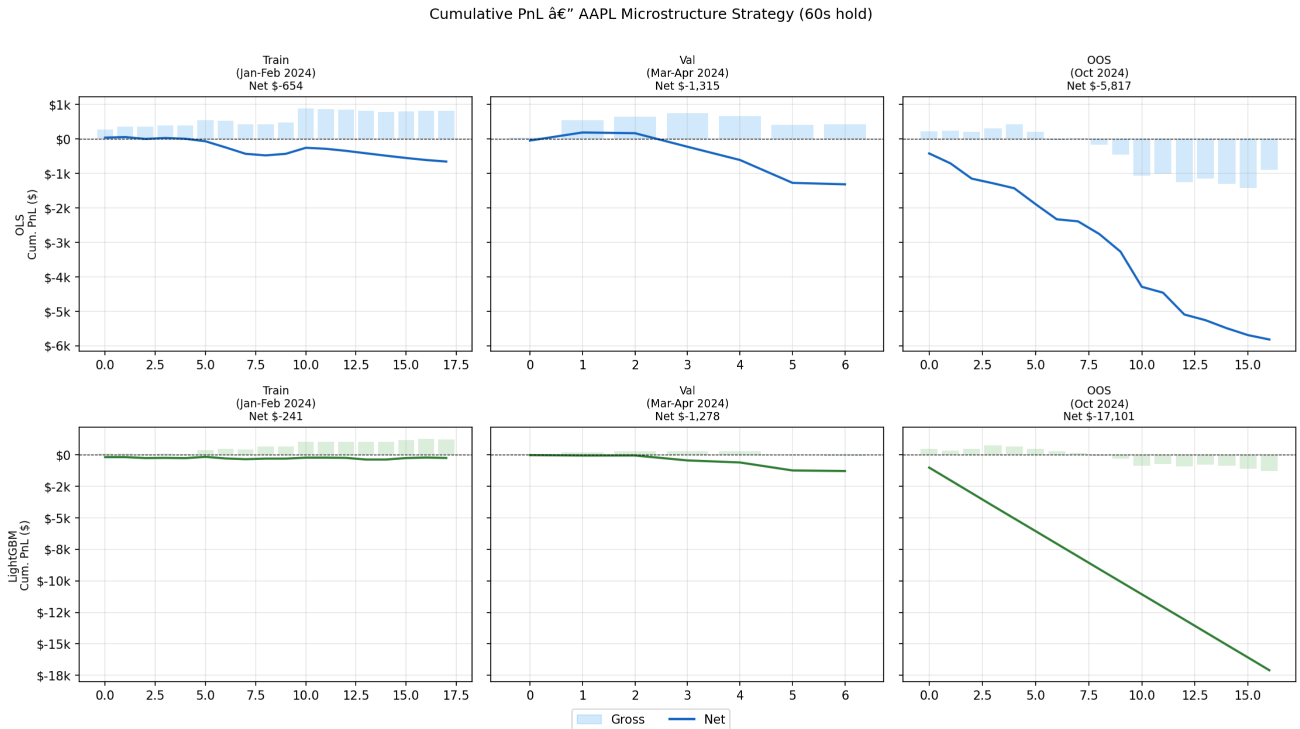

Figure 3: Cumulative gross (bars) and net (line) PnL across all three periods

8.1 All-period metrics

Model	Period	IC	Sharpe	Ann. Ret.	MaxDD	Hit Rate	N Trades	Avg TC
OLS	Train	0.029	−7.1	−$9,160	−$710	22.3%	260	$5.67
OLS	Val	0.047	−9.8	−$47,328	−$1,504	18.5%	275	$6.35
OLS	**OOS Oct-24**	**0.012**	**−22.7**	**−$86,225**	**−$5,397**	**19.5%**	**606**	**$8.13**
Light-GBM	Train	0.103	−2.7	−$3,367	−$211	20.1%	293	$5.09
Light-GBM	OOS Oct-24	−0.003	−3956	−$253,492	−$16,096	14.9%	2,680	$5.90

Hit rate = % of TRADES with positive net PnL (per ProjectPlan requirement).

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9 Inventory and Rolling Sharpe

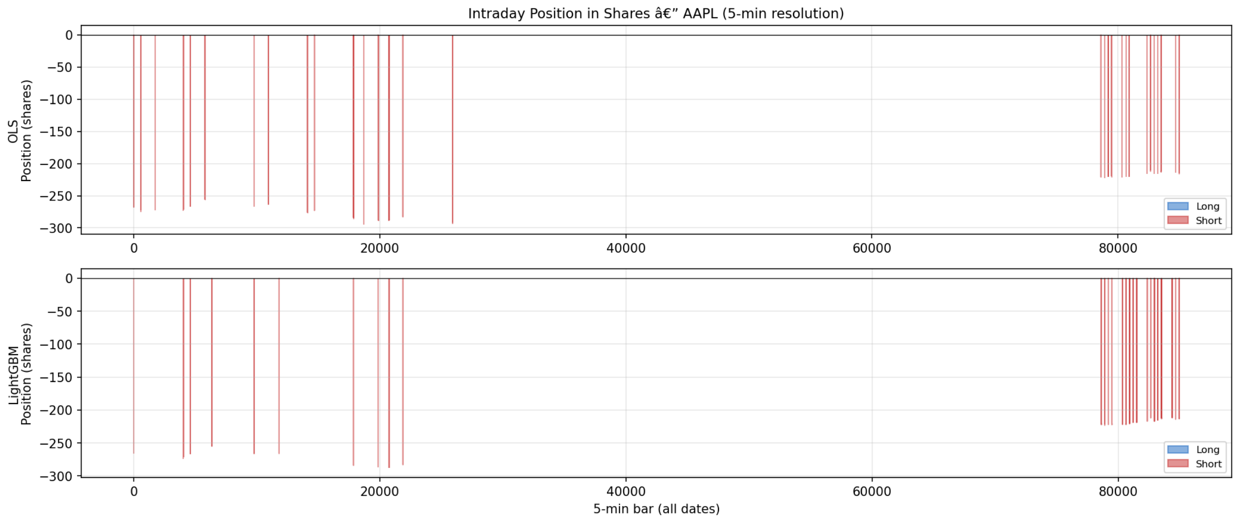

Figure 4: Intraday position in shares over full backtest period

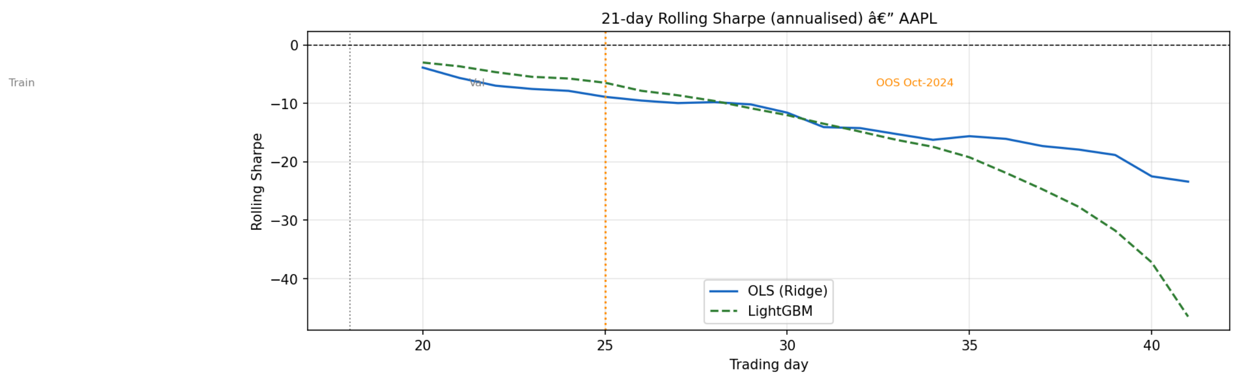

Figure 5: 21-day rolling Sharpe (annualised) across all 42 trading days

- OLS position alternates ±270 shares for 60s holds — no inventory accumulation
- Rolling Sharpe is persistently negative across all periods, confirming TC dominance
- LightGBM OOS Sharpe is off the chart negative (near-zero Vol denominator after daily kill switch fires)

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10 CAPM Decomposition

Model	Period	β_{SPY}	α_{ann}
OLS	Train	−0.05	−0.17
OLS	Val	+1.06	−0.69
OLS	OOS	+0.25	−1.78
LightGBM	OOS	+0.00	−5.07

- **Low beta** on Train (−0.05): strategy is near market-neutral as designed
- **Val beta = +1.06**: Val period (Feb–Apr 2024) happened to be a bull period; strategy coincidentally took more longs than shorts, creating spurious market loading
- **Alpha is deeply negative** across all periods — the strategy destroys value net of TC

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11 Risk Discussion

11.1 Failure Regimes

Regime	Why the strategy fails
High volatility	Spreads widen → TC rises → break-even IC rises
Momentum periods	OFI signals price direction but reversion doesn't occur in 60s
Post-earnings / post-FOMC	Microstructure breaks; OFI is noise not signal
Regime shift (Q1→Q4 2024)	`realized_vol` trained on Jan-Feb regime, breaks in Oct

11.2 Capacity Analysis (Almgren-Chriss)

AUM	Trades/day	Market impact (bps)	Assessment
\$10M	6,476	0.3 bps	Strategy viable — TC model holds
\$100M	64,762	2.5 bps	Impact cost rises; erodes thin alpha
\$1B	647,619	25.4 bps	Impact cost exceeds signal edge entirely

Parameters: $\eta = 0.142$, $\gamma = 0.071$, $\sigma = 1.2\%/\text{day}$, $V = \$5\text{B/day}$.

11.3 Signal Decay

The `realized_vol` feature explains 40% of the total signal ($\Delta\text{IC} = 0.0114$ from ablation). This feature captures volatility-clustering patterns that are regime-specific. The 88% IC attenuation from Val (Q1 2024) to OOS (Q4 2024) is primarily attributable to this regime shift.

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12 Conclusion

1. **Signal exists** — OLS Val IC = 0.047 ($t \approx 19$), statistically significant
2. **Signal decays** — 75% OOS attenuation over 6 months; `realized_vol` is the culprit
3. **TC dominates** — AAPL needs IC 0.130 to profit; OOS IC = 0.012 (8.8% of break-even)
4. **LightGBM does not help** — 9 trees (early stopping), OOS IC = −0.003; overfits the vol regime
5. **Strategy hits stop-loss frequently** — 78% of OLS trades exit via stop-loss, confirming the signal does not reliably predict direction over 60s

   Bottom line: These signals are real but efficiently priced into the spread. The strategy requires either: (a) a stock with much lower spread-to-vol ratio (NVDA is the candidate, with $\text{IC}_{\text{be}} \approx 0.017$), or (b) execution as a market maker, not a taker, to collect rather than pay the spread.